



A dual Kalman filter approach for state estimation via output-only acceleration measurements

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ABSTRACT

A dual implementation of the Kalman filter is proposed for estimating the unknown input and states of a linear state-space model by using sparse noisy acceleration measurements. The successive structure of the suggested filter prevents numerical issues attributed to unobservability and rank deficiency of the augmented formulation of the problem. Furthermore, it is shown that the proposed methodology furnishes a tool to avoid the so-called drift in the estimated input and displacements commonly encountered by existing joint input and state estimation filters. It is shown that, by fine-tuning the regulatory parameters of the proposed technique, reasonable estimates of displacements and velocities of structures can be accomplished.

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1. Introduction

This paper contributes to the problem of state estimation in the entire body of the metallic structures that undergo vibrations due to unknown input forces during their operational life, aiming at prediction of fatigue damage identification. The idea of using the estimated response of the structures for fatigue damage identification was first suggested by Papadimitriou et al. [1]; where a technique was introduced that uses the Kalman filter for estimating power spectral densities of the strain in the body of the structure thereby predicting the remaining fatigue life. To estimate the fatigue damage, a time history of the strains in the hotspot points of the structure is required. To estimate the strain in a point of interest, the displacement field around that point is needed; therefore, a reliable state estimate could lead to a reliable fatigue damage identification.

The subject of estimation of the states of a partially observed dynamic system in an stochastic frame has been studied by many scientists and there are well developed algorithms to manage both linear (e.g. the Kalman filter [2]) and nonlinear (e.g. the particle filter [3], the unscented Kalman filter [4]) state-space models. Dealing with structural systems, the states of the system are displacements and velocities of the response of the system at some points, namely degrees-of-freedom (DOF) on the structure. In practical cases, it is difficult or sometimes impossible to measure displacements and velocities of the system, hence when a knowledge of the displacements and velocities is required, a state estimation algorithm could be used to provide estimates of the whole state of the system. The Bayesian filters that exist in the literature, take advantage of the correlation between the observable part of the state of the system and the hidden part, and furnish an estimate of the whole

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state with the associated uncertainty of it. Within the context of the structural dynamics, over the continuous time the equations of motion, namely the dynamic model of the system define an intrinsic correlation between the states of the system. Dealing with discrete models, the kinematic relation between the accelerations, velocities and displacements of the system enters the dynamic model of the system and together with kinetics shape the discrete state-space model.

Within such a context, Ching and Beck [5] estimated the unknown states of a structure using incomplete output data from a structure excited by uncertain dynamic loading, to estimate the likelihood of any particular unobserved response of the structure exceeding a prescribed threshold. Hernandez [6] proposed an observer that possesses similar characteristics to the Kalman filter in the sense that it minimizes the trace of the state error covariance matrix. The main notion behind the algorithm is that the proposed observer can be implemented as a modified linear finite element model of the system, subject to collocated corrective forces proportional to the measured response. It has been used to estimate the number of threshold crossings in the bending moment history of a simulated tall vertical structure subject to turbulent wind load and fatigue damage [7]. The methodology has further been experimentally validated via a laboratory test [8]; where the measured stress at the locations of interest was compared to estimates obtained by well-established estimation methods such as Luenberger observers and the Kalman filter relying on the using a limited number of velocity measurements. Smyth and Wu [9] proposed a multi-rate Kalman filter for the online fusion of measured displacement and acceleration data sampled at different rates. The filter is designed to circumvent problems related to the integration of accelerometer or the differentiation of displacement data in situations where both these response quantities are collocated and available in different sampling rates. Gao and Lu [10] used the Kalman filter in combination with an ARX model for damage detection in linear structural systems where only accelerations at some degrees of freedom are measured. Reynders and De Roeck used the linear Kalman filter as part of the subspace identification technique for modal analysis [11] to directly estimate states from measured data without knowledge of the system model. The linear Kalman filter is also used by Bernal [12] in a damage detection scheme for linear systems based on the hypothesis test of whiteness of the innovations, taking advantage of the fact that correlations emerge when either the properties of the system or the process noise deviate from the values that the filter is formulated for. In very recent work, Bernal and Ussia [13] have proposed a sequential deconvolution method for input reconstruction in linear time invariant systems, and have presented a detailed study of the necessary conditions for identifying the input force as well as the stability conditions associated with a segmented implementation of the proposed method. Through theoretical derivations, it is shown that when the number of inputs is less than or equal to the number of observations the non-collocated inputs can be identified by performing deconvolution. Further, to facilitate continuous input reconstruction over time, a sequential implementation of the deconvolution method is proposed by authors. The method is an interesting alternative for the case where the aim is input reconstruction in the event where the input load locations are known and additionally the inputs to be reconstructed are fewer than the monitored outputs. In such a case one may solve for the inputs and then recast the problem into a standard Kalman filtering framework which would take care of the process and measurement noise involved, as well as the uncertain initial conditions. Instead, the method developed herein proposes a parallel carrying out of these tasks in one compact formulation.

For nonlinear state estimation and parameter identification in civil engineering, the extended Kalman filter (EKF) has been the de facto standard in the past mainly due to its ease of implementation, robustness and suitability for real-time applications. In recent years, however, many alternative techniques have been proposed. In a first extension for alleviating the issues that arise through linearization in the EKF, Julier and Uhlmann [4] have proposed the unscented Kalman filter (UKF), in which the evolution of the statistics of the state of the system is performed through a sampling scheme. It has been shown by Mariani and Ghisi [14] that at the price of a higher computational burden the UKF outperforms the EKF dealing with nonlinear parameter identification problems. To mitigate the issues pertinent to high computational costs, Eftekhar Azam et al. [15] proposed a parallel implementation of the UKF. Ching et al. [16] compared the performance of the EKF with that of the particle filter (PF) applied to identification of system matrices of a linear multi storey shear building. Chatzi and Smyth [17] and Eftekhar Azam et al. [18] compared the performance of the UKF and PF applied to identification of the parameters of nonlinear constitutive models. One of the advantages of the PF in comparison to the EKF is that it is applicable to highly nonlinear systems with non-Gaussian uncertainties. In turn, one major drawback of the particle filters is that when dealing with high dimensional state-space models, the computational burden of the generic particle filters increase exponentially. To alleviate this issue, recently techniques have been developed that improve the ensemble of the so-called particles. Chatzi and Smyth [19] have used evolutionary particle filters for structural health monitoring where, the use of the standard PF is combined with mutation operators to enhance the particles. Eftekhar Azam and Mariani [20] have used a hybrid extended Kalman particle filter for online damage detection of linear and nonlinear multi storey shear type buildings. In the abovementioned works, the process and observation noise covariances are always assumed as known parameters of the problem; however, in practice the covariances of the noise parameters should be appropriately estimated to ensure that an optimal prediction is furnished by the filters [21]. Moreover in practical implementations in online monitoring, the possible outliers contained in the observation process can have detrimental effects on the estimates including instability of the identified parameters. In treating this issue and for enabling online and continuous monitoring, Mu and Yuen have proposed a novel robust outlier resistant EKF for online parameter identification of linear time variant structural systems [22].

Although the joint state and parameter identification task is a subject frequently addressed in recent years, the joint identification of state and input information is a topic less treated so far in the literature. Structural systems are inherently characterized by uncertainty, relating to measurement errors, sensor noise, inefficacy of the numerical models and lack of a priori knowledge on the system and loading conditions. In this paper, the latter source of uncertainty, i.e. the lack of information

regarding the input to the system is the core of the study. In practice, one common approach is to assume the unknown input as a zero mean white Gaussian process and make use of the aforementioned Bayesian techniques for state estimation; however, in many cases this assumption is violated and therefore it may lead to major adverse effects on the accuracy of the estimations. To address this issue, a number of optimal filtering techniques in the presence of unknown input have been proposed. In a pioneering work, Kitanidis developed an unbiased minimum-variance recursive filter for input and state estimation of linear systems without direct transmission; his algorithm did not make any a-priori assumption on the input [23]. The latter filter is not globally optimal in the mean square error sense. Hsieh has proposed a new formulation of the Kitanidis filter which is more convenient for practical applications [24]. Gillijns and De Moor proposed a new filter for joint input and state estimation for linear systems without direct transmission [25]. Their filter is globally optimal in the minimum-variance unbiased sense. Later Gillijns and De Moor developed a new formulation of the aforementioned filter which included a direct transmission term in its structure [26].

In more recent years, Lourens et al. [27] have proposed an extension of the method developed in [26] to cope with the numerical instabilities that arise when the number of sensors surpasses the order of the model, i.e. when a large number of sensors is used in combination with a reduced-order model assembled from a relatively small number of modes. The modified algorithm was used to predict and estimate the input force and accelerations of a simulated steel beam, a laboratory test beam and a large scale steel bridge. It was reported that, although the algorithm provides a reasonable prediction of the accelerations, the input force estimates are affected by spurious low frequency components that must be filtered out in this case. It is worth noting, that in dealing with the joint state and parameter estimation, Chatzi and Fuggini [28] have proposed a technique to cope with the issues related to the spurious low frequency components in the displacement estimates by introducing artificial displacement measurements into the observation vector. Lourens et al. [29] have proposed an augmented Kalman filter (AKF) for unknown force identification in structural systems, and concluded that the AKF is prone to numerical instabilities due to un-observability issues of the augmented system matrix.

In this paper, a dual implementation of the Kalman filter is proposed to estimate the unknown input and states of a linear state-space model; however, the input estimation itself is a secondary goal compared to state estimation, as the objective is to estimate the fatigue damage accumulation. It is assumed that a limited number of noisy acceleration measurements are available. The successive structure of the suggested filter prevents numerical problems attributed to un-observability and rank deficiency of the AKF. Additionally, it is shown that the expert guess on the covariance of the unknown input provides a tool for avoiding the so-called drift effect in the estimated input force and displacements. The drift is linked to the integral nature of these quantities in the presence of acceleration information. The effectiveness and performance of the proposed method is ascertained via numerical analysis carried out on a shear model of a building as well as the numerical model of the Pirelli Tower [30,31], a land mark skyscraper located at Milan, Italy. It is concluded that, by fine-tuning the covariance of the fictitious process noise of the unknown input, a reasonable estimate of the state, useful for fatigue damage estimation could be accomplished.

The paper starts with a section devoted to a brief formulation of the state-space equations for linear dynamical systems. The next section introduces the dual scheme by use of the Kalman filter for estimation of both the unknown input and state of linear state-space models and is followed by a section on the numerical comparison of the dual Kalman filter, the augmented Kalman filter [29] and the filter proposed by Gillijn and De Moor (GDF) [26]. The paper is concluded by a section on the numerical investigation of the performance of the proposed algorithm when applied to both input and state estimation of the Pirelli Tower, located in Milan, Italy.

2. Mathematical formulation of the problem

A linear structural dynamics problem is typically formulated using the following continuous time second order differential equation:

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{f}(t) = \mathbf{S}_p \mathbf{p}(t) \quad (1)$$

where $\mathbf{u}(t) \in \mathbb{R}^n$ denotes the displacement vector and $\mathbf{K}$, $\mathbf{C}$ and $\mathbf{M} \in \mathbb{R}^{n \times n}$ stand for the stiffness, damping and mass matrix, respectively. $\mathbf{f}(t) \in \mathbb{R}^n$ is the excitation force, which herein is presented as a superposition of time histories $\mathbf{p}(t) \in \mathbb{R}^m$ that are influencing some degrees-of-freedom on the structure as indicated via the influence matrix $\mathbf{S}_p \in \mathbb{R}^{n \times m}$.

In practice, when dealing with fine resolution finite element (FE) models, the dimension of the state vector in Eq. (1) may become relatively large; nonetheless the dynamics of the system could effectively be captured by a significantly smaller number of modes. To suppress the computational costs associated with the large FE models, Eq. (1) is projected in the subspace spanned by a limited number of the undamped eigenmodes of the system. In this regard, consider the eigenvalue problem corresponding to Eq. (1)

$$\mathbf{K}\Phi = \mathbf{M}\Phi\Omega^2 \quad (2)$$

Transforming the coordinate system of Eq. (1) via the following mapping:

$$\mathbf{u}(t) = \Phi \mathbf{z}(t) \quad (3)$$

where $\mathbf{z}(t) \in \mathbb{R}^m$, $\Phi \in \mathbb{R}^{n \times m}$, and pre multiplying by Φ^T imposing the mass normalization condition $\Phi^T \mathbf{M} \Phi = \mathbf{I}$, considering $\Phi^T \mathbf{K} \Phi = \Omega^2$ and assuming the damping is proportional, Eq. (6) can be rewritten

$$\ddot{\mathbf{z}}(t) + \Gamma \dot{\mathbf{z}}(t) + \Omega^2 \mathbf{z}(t) = \Phi^T \mathbf{f}(t) = \Phi^T \mathbf{S}_p \mathbf{p}(t) \quad (4)$$

where the components of j th entry of the diagonal damping matrix Γ are of the form $2\xi_j \omega_j$, in which ξ_j stands for the relevant modal damping ratio. Apparently, a truncated modal space could be substituted in Eq. (4).

The aforementioned equations can be discretized in time to constitute a state-space equation, and in so doing the following state vector is introduced:

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{u}(t) \\ \dot{\mathbf{u}}(t) \end{bmatrix}$$

Consequently, Eq. (1) can be written in the following form to define the process equation:

$$\dot{\mathbf{x}}(t) = \mathbf{A}_c \mathbf{x}(t) + \mathbf{B}_c \mathbf{p}(t) \quad (5)$$

where the system matrices are:

$$\mathbf{A}_c = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{M}^{-1} \mathbf{K} & -\mathbf{M}^{-1} \mathbf{C} \end{bmatrix}$$

$$\mathbf{B}_c = \begin{bmatrix} 0 \\ -\mathbf{M}^{-1} \mathbf{S}_p \end{bmatrix}$$

Regarding the measurement equation let us consider the most general case by assuming that a combination of the displacements, velocities and accelerations can be measured. Hence, the measurement vector $\mathbf{d}(t)$ assumes the following form:

$$\mathbf{d}(t) = \begin{bmatrix} \mathbf{S}_d & 0 & 0 \\ 0 & \mathbf{S}_v & 0 \\ 0 & 0 & \mathbf{S}_a \end{bmatrix} \begin{bmatrix} \mathbf{u}(t) \\ \dot{\mathbf{u}}(t) \\ \ddot{\mathbf{u}}(t) \end{bmatrix} \quad (6)$$

where $\mathbf{S}_d$, $\mathbf{S}_v$ and $\mathbf{S}_a$ are the selection matrices of appropriate dimension for the displacements, velocities and accelerations, respectively. By using equation of motion, Eq. (6) could be transformed into state-space form

$$\mathbf{d}(t) = \mathbf{G}_c \mathbf{x}(t) + \mathbf{J}_c \mathbf{p}(t) \quad (7)$$

where the output influence matrix and the direct transmission matrix are

$$\mathbf{G}_c = \begin{bmatrix} \mathbf{S}_d & 0 \\ 0 & \mathbf{S}_v \\ \mathbf{S}_a \mathbf{M}^{-1} \mathbf{K} & \mathbf{S}_a \mathbf{M}^{-1} \mathbf{C} \end{bmatrix}$$

$$\mathbf{J}_c = \begin{bmatrix} 0 \\ 0 \\ \mathbf{S}_a \mathbf{M}^{-1} \mathbf{S}_p \end{bmatrix}$$

Recombining Eqs. (5) and (7) through use of the relevant matrices, results into the full order state-space equations that are required to implement the input and state estimation algorithm. In case a reduced order state-space model is needed, a truncated eigenvector space must be substituted in Eq. (4); hence the following variable transformation would be necessary:

$$\mathbf{x}(t) = \begin{bmatrix} \Phi_r & 0 \\ 0 & \Phi_r \end{bmatrix} \zeta(t)$$

where $\zeta(t)$ is the reduced modal state vector

$$\zeta(t) = \begin{bmatrix} \mathbf{z}(t) \\ \dot{\mathbf{z}}(t) \end{bmatrix}$$

The reduced modal state-space equation in continuous time will have the following form:

$$\dot{\zeta}(t) = \mathbf{A}_c \zeta(t) + \mathbf{B}_c \mathbf{p}(t) \quad (8)$$

$$\mathbf{d}(t) = \mathbf{G}_c \zeta(t) + \mathbf{J}_c \mathbf{p}(t) \quad (9)$$

while the relevant system matrices read

$$\mathbf{A}_c = \begin{bmatrix} 0 & \mathbf{I} \\ -\Omega^2 & -\Gamma \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} 0 \\ -\Phi_r^T \mathbf{S}_p \end{bmatrix}, \quad \mathbf{G}_c = \begin{bmatrix} \mathbf{S}_d \Phi_r & 0 \\ 0 & \mathbf{S}_v \Phi_r \\ \mathbf{S}_a \Phi_r \Omega^2 & \mathbf{S}_a \Phi_r \Gamma \end{bmatrix}, \quad \mathbf{J}_c = \begin{bmatrix} 0 \\ 0 \\ \mathbf{S}_a \Phi_r \Phi_r^T \mathbf{S}_p \end{bmatrix}$$

To discretize Eqs. (8) and (9), the sampling rate is denoted by $1/\Delta t$ and the discrete time instants are defined at $t_k = k \Delta t$, for $k = 1, \dots, N$. The discrete state-space equation can be expressed by the following notation:

$$\zeta_{k+1} = \mathbf{A}\zeta_k + \mathbf{B}\mathbf{p}_k \quad (10)$$

$$\mathbf{d}_k = \mathbf{G}\zeta_k + \mathbf{J}\mathbf{p}_k \quad (11)$$

where $\mathbf{A} = \mathbf{e}^{\mathbf{A}_c \Delta t}$, $\mathbf{B} = [\mathbf{A} - \mathbf{I}]\mathbf{A}_c^{-1}\mathbf{B}_c$, $\mathbf{G} = \mathbf{G}_c$ and $\mathbf{J} = \mathbf{J}_c$. It is noteworthy that, in this study the $\mathbf{B}$ and $\mathbf{J}$ matrices are converted from continuous time to discrete via a zero-order-hold (ZOH) assumption, which assumes a constant inter-sample behavior for the input. It should be noted at this point, that Bernal [32] has carried out a thorough study of other more realistic assumptions on the inter-sample behaviors of the input for dynamic systems and concluded that a Dirac comb impulse assumption can significantly improve the discretization accuracy. A further analysis of this issue lies beyond the scope of this article and the interested reader is referred to [32]. The ZOH assumption is adopted herein for the further purpose of allowing the direct cross-comparison of this work with the methodologies preceding it as outlined in the introductory section, which also relied on the ZOH assumption.

3. Dual Kalman filter for input and state estimation

Consider the following discrete time state-space equation:

$$\zeta_{k+1} = \mathbf{A}\zeta_k + \mathbf{B}\mathbf{p}_k + \mathbf{v}_k^\zeta \quad (12)$$

$$\mathbf{d}_k = \mathbf{G}\zeta_k + \mathbf{J}\mathbf{p}_k + \mathbf{w}_k \quad (13)$$

where $\mathbf{v}_k^\zeta$ is the process noise assumed, zero-mean, white with covariance $\mathbf{Q}^\zeta$, and $\mathbf{w}_k$ is the zero mean, white, measurement noise of covariance $\mathbf{R}$. The problem at hand is to estimate the unknown input $\mathbf{p}_k$ and the hidden or partially observed state ζ_k of the system using the noisy observations $\mathbf{d}_k$ in an online fashion. In doing so, a dual implementation of the Kalman filter is proposed in this section. The proposed scheme could be divided into two stages, with the Kalman filter pertaining to both stages. At each time iteration, a fictitious process equation serving for calibration of the input force is introduced

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k^p \quad (14)$$

where $\mathbf{v}_k^p$ is a zero mean white Gaussian process with an associated covariance matrix $\mathbf{Q}^p$. Now, assume that an estimation of the state at time t_k is available; by using Eqs. (13) and (14), a new state-space equation can be obtained, where, the observed quantity is $\mathbf{d}_k$, the new state is $\mathbf{p}_k$ and the actual sought-for state ζ_k plays the role of a known input to the system:

$$\begin{aligned} \mathbf{p}_{k+1} &= \mathbf{p}_k + \mathbf{v}_k^p \\ \mathbf{d}_k &= \mathbf{G}\zeta_k + \mathbf{J}\mathbf{p}_k + \mathbf{w}_k \end{aligned}$$

Through implementation of the Kalman filter, an online estimation of $\mathbf{p}_{k+1}$ could be obtained. Then, once the estimation of $\mathbf{p}_{k+1}$ is performed, it can in a next step be substituted in Eqs. (12) and (13), and a subsequent Kalman filter implementation could be used for estimating ζ_{k+1} . The general scheme is described in detail in Table 1.

At this point, it is worth noting that, the procedure needs a-priori information on expected value and covariance of the state and input at time t_0 . Moreover, similar to the AKF, the value of the process noise $\mathbf{Q}^p$ for Eq. (14) must be properly chosen so that an accurate estimate of the unobserved state and the unknown input could be achieved. In the jargon of system identification, the covariance noise of the sought-for parameter is sometimes called the tuning knob of the system, and typically heuristic and ad-hoc guidelines are prescribed for a proper adjustment [33,34]. Methods relying on the use of Bayesian techniques, maximizing the likelihood of measurements with respect to the noise parameters have also recently been proposed [35]. It is additionally, helpful to clarify the nature of the influence of the covariance matrices $\mathbf{Q}^\zeta$, $\mathbf{Q}^p$, $\mathbf{R}$. The process noise covariance matrices reveal the confidence put on the utilized model of the system. The lower this is, the more accurate the model is considered to be. The observation noise covariance reveals the confidence put in the acquired measurements. The lower this is, the tighter the estimator is forced to fit the recorded data.

In what follows, by use of an illustrative numerical example, the performance of the proposed algorithm, i.e. the Dual Kalman filter (DKF) formulation, is evaluated against the GDF, which is deemed as the most stable attempt so far in addressing the joint input and state estimation problem. It will be shown that once the covariance of the fictitious process equation of the input force in DKF is tuned properly, the so-called drift in the estimates of the input force and the displacements is disappeared. Moreover, it is shown that the successive structure of the DKF does not trigger unobservability issues reported on the application of AKF [29].

4. Simulated example

To assess the performance of the proposed algorithm, an 8 DOF shear building (see Fig. 1) with system properties introduced in [36] is adopted, where the value of the mass of each floor is assumed to be 625 tones, and the inter-storey stiffness of each floor is equal to 10^9 kgf/m. Additionally, the modal damping ratio of each mode is assumed to be 2%.

Throughout the numerical analysis section, it is assumed that only accelerations of the response of the structure at the storey levels are available. This is the common case in structural dynamics; in practice the displacements and velocities are

Table 1

The general scheme of the two-stage Kalman filter-based input and state estimation algorithm.

– Initialization at time t_0 :

$$\begin{aligned}\hat{\mathbf{p}}_0 &= \mathbb{E}[\mathbf{p}_0] \\ \mathbf{P}_0^p &= \mathbb{E}[(\mathbf{p}_0 - \hat{\mathbf{p}}_0)(\mathbf{p}_0 - \hat{\mathbf{p}}_0)^T] \\ \hat{\zeta}_0 &= \mathbb{E}[\zeta_0] \\ \mathbf{P}_0 &= \mathbb{E}[(\zeta_0 - \hat{\zeta}_0)(\zeta_0 - \hat{\zeta}_0)^T]\end{aligned}$$

– At time t_k , for $k = 1, \dots, N_t$:

• Prediction stage for the input:

1. Evolution of the input and prediction of covariance input:

$$\begin{aligned}\mathbf{p}_k^- &= \mathbf{p}_{k-1} \\ \mathbf{P}_k^{p-} &= \mathbf{P}_{k-1}^p + \mathbf{Q}^p\end{aligned}$$

• Update stage for the input:

2. Calculation of Kalman gain for input:

$$\mathbf{G}_k^p = \mathbf{P}_k^{p-} \mathbf{J}^T (\mathbf{J} \mathbf{P}_k^{p-} \mathbf{J}^T + \mathbf{R})^{-1}$$

3. Improve predictions of input using latest observation:

$$\begin{aligned}\hat{\mathbf{p}}_k &= \mathbf{p}_k^- + \mathbf{G}_k^p (\mathbf{d}_k - \mathbf{G} \hat{\zeta}_{k-1} - \mathbf{J} \mathbf{p}_k^-) \\ \mathbf{P}_k^p &= \mathbf{P}_k^{p-} - \mathbf{G}_k^p \mathbf{J} \mathbf{P}_k^{p-}\end{aligned}$$

• Prediction stage for the state:

4. Evolution of state and prediction of covariance of state:

$$\begin{aligned}\zeta_k^- &= \mathbf{A} \hat{\zeta}_{k-1} + \mathbf{B} \hat{\mathbf{p}}_k \\ \mathbf{P}_k^- &= \mathbf{A} \mathbf{P}_{k-1} \mathbf{A}^T + \mathbf{Q}^\zeta\end{aligned}$$

• Update stage for the state:

5. Calculation of Kalman gain for state:

$$\mathbf{G}_k^\zeta = \mathbf{P}_k^- \mathbf{G}^T (\mathbf{G} \mathbf{P}_k^- \mathbf{G}^T + \mathbf{R})^{-1}$$

6. Improve predictions of state using latest observation:

$$\begin{aligned}\hat{\zeta}_k &= \zeta_k^- + \mathbf{G}_k^\zeta (\mathbf{d}_k - \mathbf{G} \zeta_k^- - \mathbf{J} \hat{\mathbf{p}}_k) \\ \mathbf{P}_k &= \mathbf{P}_k^- - \mathbf{G}_k^\zeta \mathbf{G} \mathbf{P}_k^-\end{aligned}$$

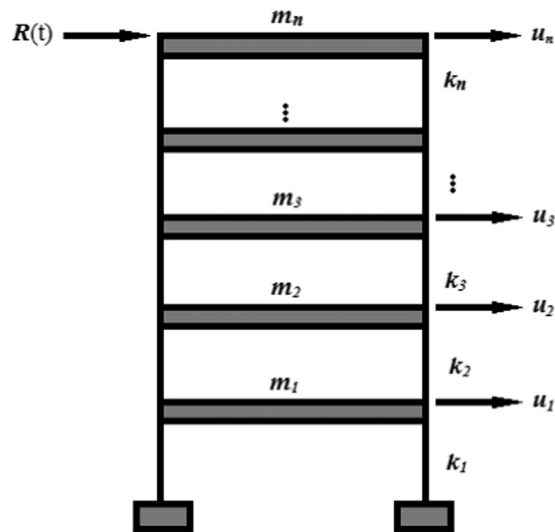
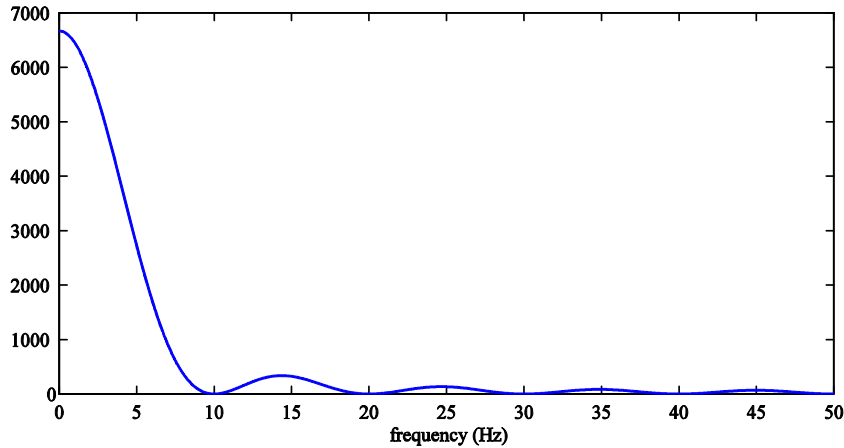
**Fig. 1.** Schematic view of a shear-type building.

Table 2

The eight undamped natural frequencies of the structure.

Vibration mode index	1	2	3	4	5	6	7	8
Natural frequency (Hz)	1.17	3.48	5.67	7.67	9.41	10.83	11.87	12.52

**Fig. 2.** Spectrum of the impulse excitation applied to the shear building.

difficult, or even sometimes impossible to measure. Therefore, the problem lies in estimating the displacements and velocities of all stories of the structure by using noisy observations acquired from acceleration sensors. The optimization of the spatial distribution of the sensors [37] is not within the scope of this research but would be an interesting issue to explore in a follow-up investigation.

The study presented herein is performed by using simulated, noise contaminated data: to account for measurement errors, a zero mean white Gaussian noise is added to the simulated acceleration time histories of the structure. By varying the level of the covariance of the added noise different signal-to-noise ratios are obtained. The undamped natural frequencies of the system are reported in the Table 2.

The common trend in the state-of-the-art algorithms for unknown input and state estimation available in the literature (e.g. [23–26]), is to avoid using any a-priori knowledge on the statistics of the input force, in an attempt to render the online estimation more practical.

To make a comparison of the performances of different schemes, the mean squared error of the observed quantities of interest as well as the time histories of the sought-for states are cross-compared. In the following examples, it is assumed that an impact load is applied to the 8th floor and a single noisy acceleration observation from the 8th floor of the shear building is available. The impact load has a triangular time history: it starts at $t=2$ s of the analysis, it takes 0.1 s to reach to its peak value which amounts 10^6 kgf and in another 0.1 s it descends back to zero. The spectrum of the applied force is shown in Fig. 2.

The standard deviation of the noise process is equal to 0.01 m/s^2 (approximately 1.25% of peak acceleration). The estimation of the acceleration, velocity and displacement time histories furnished by the Kalman filter algorithm are shown in Fig. 3, for the case of known input; aiming at showing the excellent performance of the Kalman filter in state estimation in the case that input is already known. In all the simulations the model deployed in the algorithms is assumed to be accurate, hence the process noise is set to a small value $\mathbf{Q}^\zeta = 10^{-20} \times \mathbf{I}$; henceforth $\mathbf{I}$ is an identity matrix of appropriate dimension. Concerning the initial values of the covariance of the state, the value used for the process noise is adopted, moreover, the system is supposed to be at rest in the beginning of the simulations therefore the expected values for the initial conditions are assumed to be zero.

Dealing with the state estimation for systems with unknown input, GDF does not include any a-priori assumption on the statistics of the input to the system. However, to furnish the estimates of the input and states of the system, AKF and DKF need the initial values of the expected value and the covariance of input force. The value of the covariance of the input force, which plays the role of the regularization parameter strongly influences the quality of the estimates of the Bayesian filters. In this study, we make recourse to L -curve in order to calibrate the values of the process noise for the input estimation. Fig. 4 shows the MSE of the acceleration time history of the 8th floor when AKF and DKF are adopted: the ordinate of the figure stands for the mean value of the square of novelty term in the Kalman filter $\sum \|\mathbf{d}_k - \mathbf{G}\zeta_k^- - \mathbf{J}\hat{\mathbf{p}}_k\|_2^2/N_t$ and the abscissa denotes the corresponding values for the covariance of the fictitious process noise. It is seen that, as reported in the [29] concerning AKF, the plots does not seem like a perfect L -curve; nevertheless, the value of $\mathbf{Q}^p = 10^9 \times \mathbf{I}$ could be intuitively chosen for AKF and DKF.

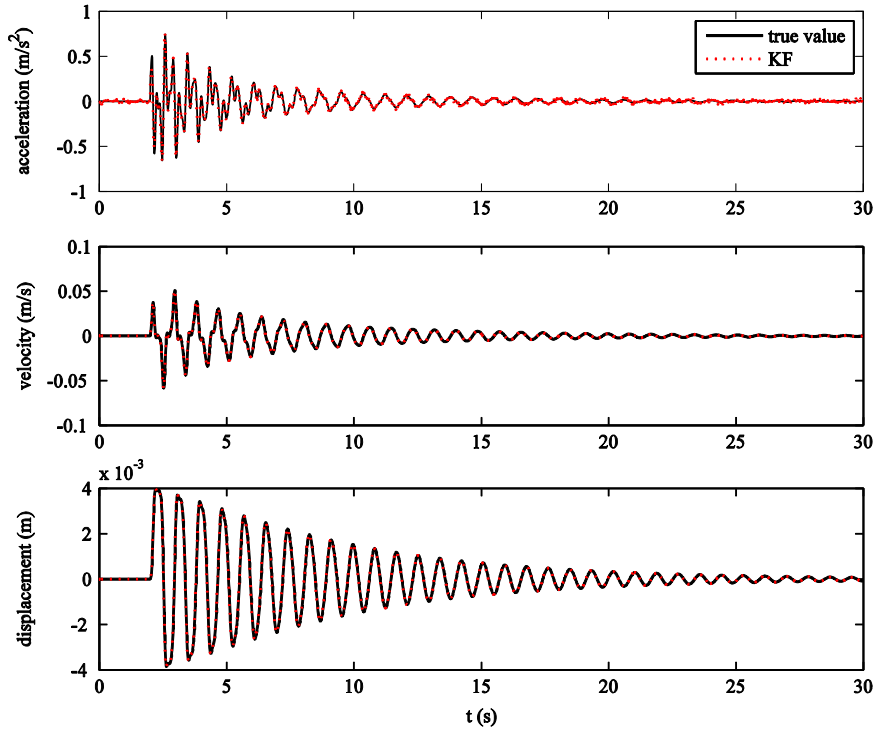


Fig. 3. Time histories estimated by the Kalman filter at DOF8 in the case of known input acceleration (top), velocity (middle), and displacement (bottom).

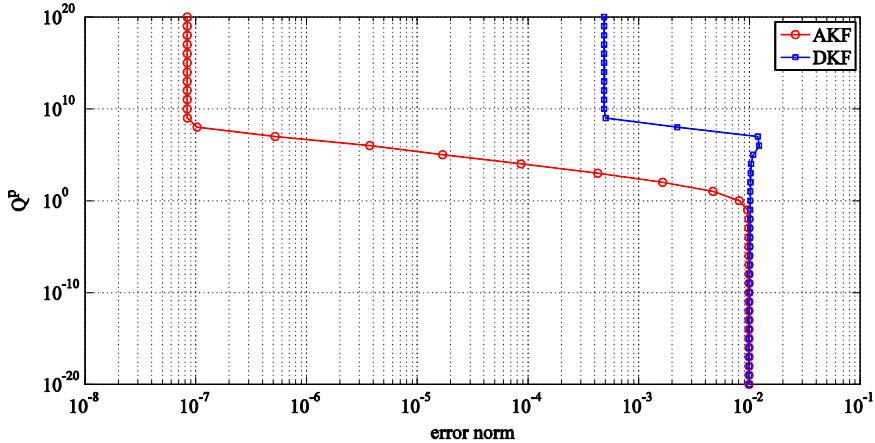


Fig. 4. L-curve for the state and input identification of the shear building using AKF and DKF.

In Fig. 5, the time histories estimated by the AKF, GDF and DKF are presented for the case of an unknown input. It is observed that the AKF, GDF and DKF are able to deliver a highly accurate estimation of the acceleration signal. However, in the velocity signal, it is noted that the accumulation of the integration errors affects the quality of the estimates provided by the GDF filter. Moreover, it can be observed that the accumulation of the double integration errors imposes a diverging estimation of the integral quantity of displacement for the GDF; the same trend, namely drift due to low frequency components stemming from double integration errors is also reported in [27]. The other way around, by reviewing the aforementioned figure, it is seen that in the estimates provided by the AKF and DKF there is no low frequency drift effect caused by the accumulation of the integration and double integration errors in velocity and displacement time histories. Moreover, it is seen that, the DKF could yield a better match with the target value of the displacement and velocity. Similar results are obtained for the rest of model DOFs.

The input force estimation time histories are shown in Fig. 6. It is seen that, likewise to the displacement estimations, the input force estimation provided by GDF suffers from a severe low frequency drift compared to the original/target value. This is due to the fact that the GDF is designed to minimize the MSE of the estimations $\sum \|\mathbf{d}_k - \mathbf{G}\zeta_k - \mathbf{J}\mathbf{p}_k\|_2^2 / N_t$ of the observed states [25], and as it turns out a false prediction for the input force estimation could provide a lower discrepancy from the

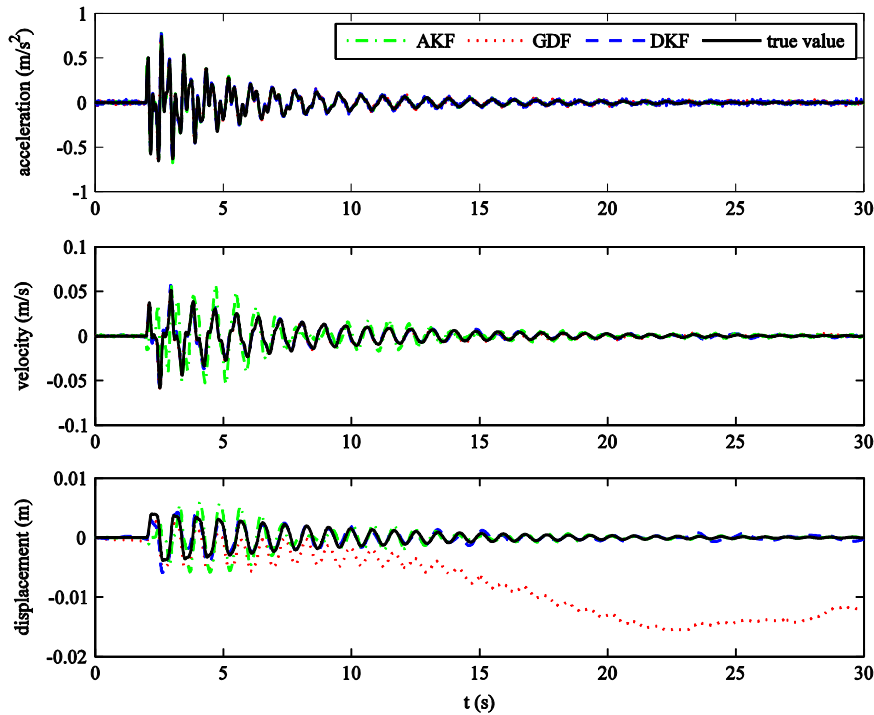


Fig. 5. Time histories estimated at DOF 8 by the AKF, GDF and DKF in the case of unknown input acceleration (top), velocity (middle), and displacement (bottom).

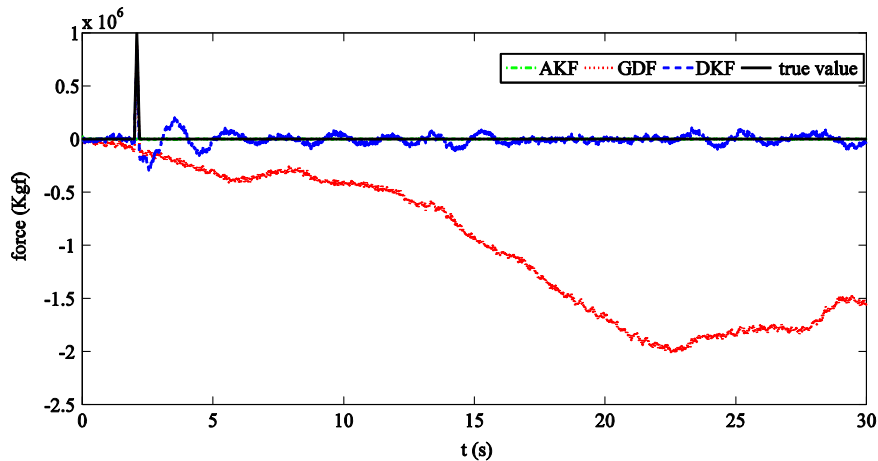


Fig. 6. The input force time history estimated by AKF, GDF and DKF in the case of the unknown input.

target value, as seen in Fig. 7 which shows the results of MSE for 100 simulations. Hence the filter is biased to the wrong estimation of displacement, velocity and input force. This is also dependent upon the numerical integration scheme chosen for the conversion for the continuous system to a discrete one, which however in the framework presented herein needs to be an online one. Therefore, we wish to avoid offline processes such as high-pass filtering and detrending [38].

The same trend is seen in the estimates provided by GDF in Fig. 6, that presents the input force time histories of the estimations by the three filters. Concerning the estimates provided by DKF, it is seen that, although the time history of input force estimation is not matching the correct value during the analysis, it is converging to the true value in the steady state. Initially, the DKF is able to capture the peak of the impulse, however, as the simulation proceeds the rather large value used for the process noise causes large oscillations around the true value of the input force. In the latter case, the AKF did not provide reasonable estimates for the input force due to unobservability of the input force in the augmented formulation.

The effect of the noise covariance in the convergence of the filter algorithm is hard to formalize. This is due to the fact, that this contribution is more of numerical nature rather than a purely theoretical one, in contrast to other concepts that can be treated on a more theoretical level, such as for instance that of observability. In quantifying this effect therefore, an

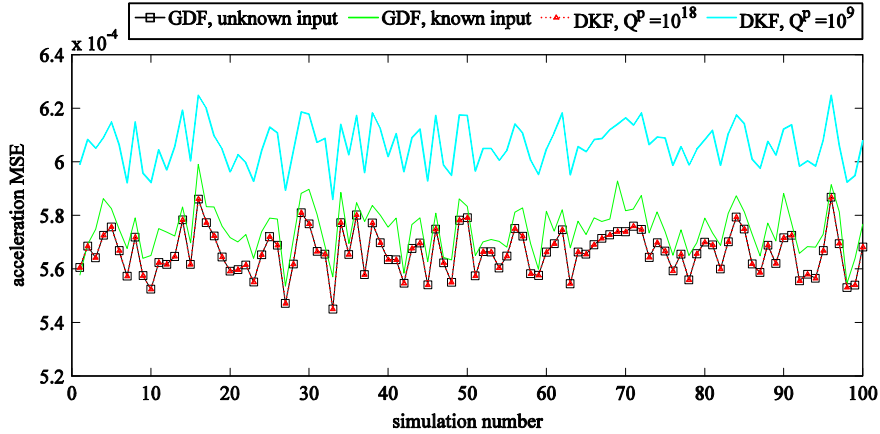


Fig. 7. Monte Carlo simulations of MSE of acceleration time history at DOF 8 estimated by GDF and DKF with respect to its true value.

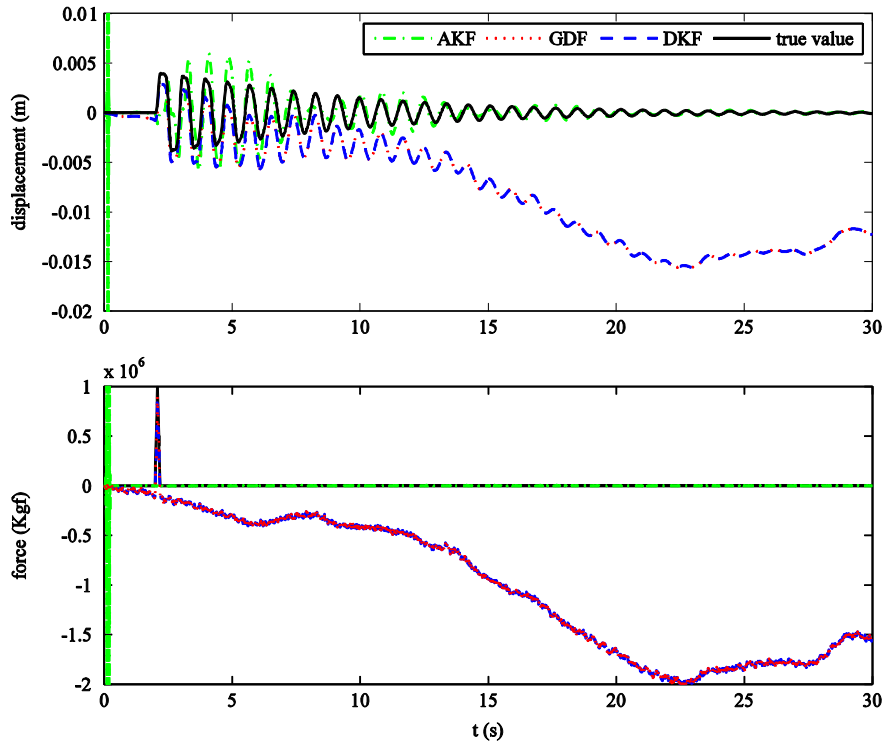


Fig. 8. Time histories estimated by AKF, GDF and DKF when $Q^p = 10^{18} \times I$.

exemplary demonstration approach is followed herein, where we demonstrate the effect of tuning this covariance for different levels. An interesting observation in Fig. 7, for instance, it is found that when increasing the value of the covariance of the fictitious process equation for the estimation of the input force to $1 \times 10^{18} \text{ kgf}^2$ in the DKF, the results of the GDF can be reproduced. The increased value of the fictitious noise provides the filter with more freedom to adjust itself with the smaller MSE for the estimated acceleration signal. Indeed, in this case the MSE of the estimated acceleration is equal to the MSE obtained by GDF. Fig. 8, present the results of the displacement and input force estimation by the AKF, GDF and DKF strategy by increasing the process noise for input estimation to $1 \times 10^{18} \text{ kgf}^2$. As it is seen in, the drift effect which was already observed in the estimates yielded by GDF affects the estimations of DKF.

In what preceded, the performance of AKF, GDF and DKF were assessed for impulse excitation of the test structure. In the remainder of this section the harmonic and seismic excitations will be assessed as well. Regarding harmonic excitation, a constant amplitude sinusoidal excitation $a_m \sin 2\pi\omega t$ is applied to the last floor of the building, where a_m denotes the amplitude of the force and ω stands for the associated frequency. In the current study, the $\omega = 0.5$ and $a_m = 5 \times 10^7 \text{ kgf}$.

The observed process is the simulated acceleration time history of the last floor contaminated with a zero mean white Gaussian process, featuring a standard deviation equal to 0.032 m/s^2 (approximately 2.5% of peak acceleration).

Figs. 9 and 10 show the estimated displacement and input force time histories furnished by AKF, GDF and DKF. The values for the diagonal components of $\mathbf{P}_0$, $\mathbf{Q}^\xi$ and $\mathbf{R}$ are set to 10^{-20} , 10^{-20} and 10^{-3} , respectively, whereas the $\mathbf{Q}^p$ used in DKF is set to 10^{12} kgf^2 . Concerning the AKF, the variations of the latter parameter does not result in an improved estimation, hence the results shown are relevant to same value used for DKF.

It is seen that, in the case that only accelerations are assumed to be observed, the un-observability issues affect the quality of estimates furnished by AKF in the case of the forced vibration caused by a harmonic excitation. On the other side, in presence of the noisy measurements, the estimates provided by GDF are affected by the so-called drift. However, the DKF seems to provide reasonable estimates of input and displacement time histories.

Next, the performances of the algorithms are assessed when dealing with seismic excitations. In so doing, the accelerogram of the May 6-1976 M_w 6.4, Friuli is considered. The earthquake is applied to the base of the structure; however, their inertial resultant is applied to each floor. In latter case, the dynamic governing equation of motion of the structure takes the following form:

$$\mathbf{M}\ddot{\mathbf{u}}_r(t) + \mathbf{C}\dot{\mathbf{u}}_r(t) + \mathbf{K}\mathbf{u}_r(t) = \mathbf{f}(t) = -\mathbf{M}\ddot{\mathbf{u}}_g(t)\mathbf{S}_p \quad (15)$$

where the subscript r refers to a coordinate system which is fixed with respect to the ground movements whereas the subscript g refers to the coordinate system which moves according to seismic ground motions. The matrix $\mathbf{S}_p$ in this case applies the accelerations to all floors of the structure.

The $\mathbf{P}_0$, $\mathbf{Q}^\xi$ and $\mathbf{R}$ are set to 10^{-10} , 10^{-10} and 10^{-3} , respectively, whereas the $\mathbf{Q}^p$ used in DKF is set to 10^{10} kgf^2 . Concerning the AKF, the variations of the latter parameter does not result in an improved estimation, hence the results shown are relevant to same value used for DKF. The observation noise $\mathbf{R}$ is supposed to be a-priori known from the manufacturer of the measurement instrument. In this study the value of it is set to the one used to produce the experimental data; nonetheless, setting a value twice larger or one half of the correct one does not lead to visible changes in the quality of estimations.

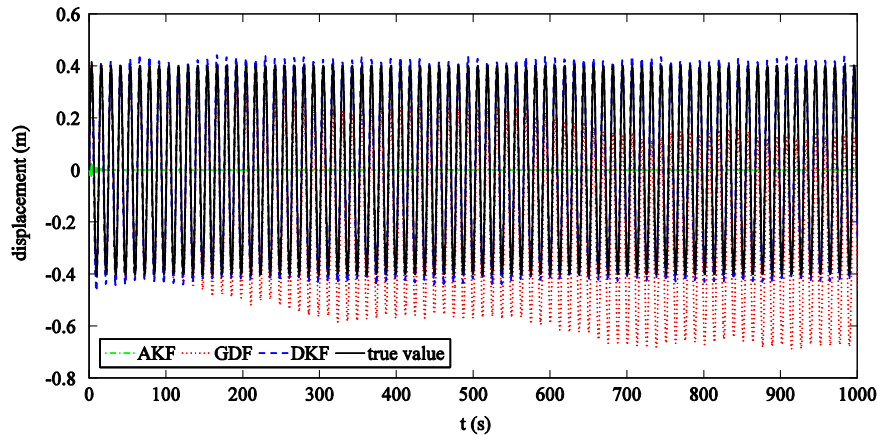


Fig. 9. Displacement time history of the shear building estimated by the AKF, GDF and DKF in the case of unknown sinusoidal excitation.

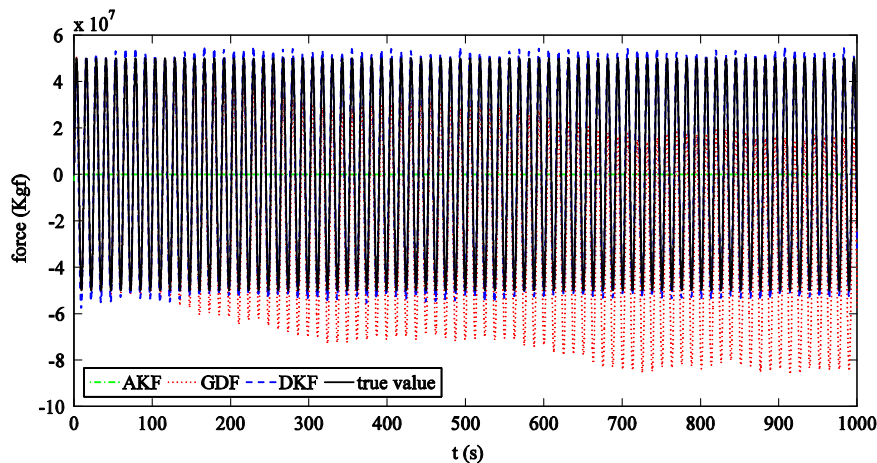


Fig. 10. Input force time history of the shear building estimated by the AKF, GDF and DKF in the case of unknown sinusoidal excitation.

Fig. 11 shows the estimated displacement time histories furnished by AKF, GDF and DKF when only the accelerations of the last floor are taken into account. It is seen that, the accumulation of the integration errors affects the estimations of GDF and AKF; however, the DKF furnishes reasonable estimates of the displacement time history. Fig. 12 shows the input force estimation provided by the AKF, GDF and DKF. It is seen that, regarding AKF the input force is unobservable in this case, whereas the DKF does not provide an accurate estimation of the unknown input force. The estimates provided by GDF are not accurate as well. In addition, in the GDF estimates low frequency components are visible. It is seen that similar to the AKF and GDF, the DKF does not provide an accurate estimation of the unknown input force; however, by a tuning of the process noise, it is possible to achieve a reasonable estimation of the input force at the cost of a less accurate displacement time history estimation. This paper mainly deals with the state estimation in situations where the input is unknown, therefore the emphasis is to obtain an accurate estimation of the displacement and velocity time histories. That is, the filters are tuned for providing an optimal estimation for the states rather than the input force.

Next, the performances of the AKF, GDF and DKF are assessed for the case that four independent white noises are applied to DOFs 2, 4, 6 and 8. In latter case, it is assumed that the noise contaminated acceleration time history of the DOF 3 constitute the observation process. It should be mentioned that, in this case there are more independent inputs than the outputs; when the aforementioned condition dominates the set up the deconvolution method fails to furnish estimates of the input process. Moreover, the observation process is chosen so that non-collocated input estimation case can be examined. Fig. 13 demonstrates the results of the state and acceleration estimation at DOF 5; it is observed that the filters furnish estimates of the states and acceleration. However, GDF still suffers from the low frequency drift. In this example, the AKF estimates are not affected by the unobservability issues, and match the accuracy of estimates furnished by DKF. It should be mentioned that the filters did not provide any estimate of the non-collocated inputs. To examine the performances of the three filters when collocated set ups are dealt, a case in which DOFs 5 and 8 comprise the observation process is studied. Fig. 14 shows the state and acceleration estimation at DOF 5, it is observed that the estimates of the three methods is qualitatively similar to the solely non-collocated case. Fig. 15 demonstrates the input estimations provided by the

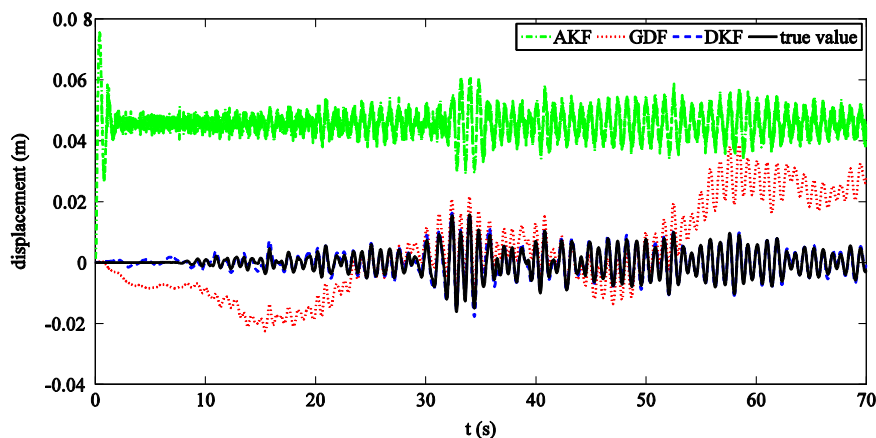


Fig. 11. Estimated displacement time history at DOF 8 of the shear building by AKF, GDF and DKF for the Friuli earthquake.

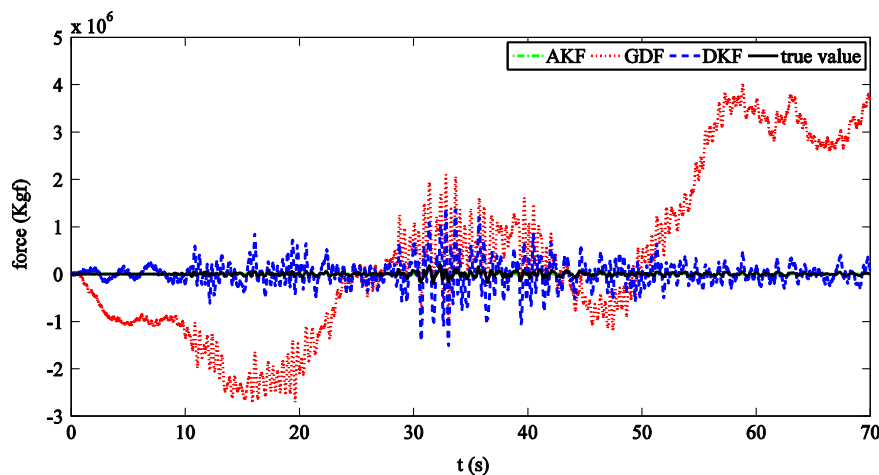


Fig. 12. Estimated input force time history of the shear building by AKF, GDF and DKF for the Friuli earthquake.

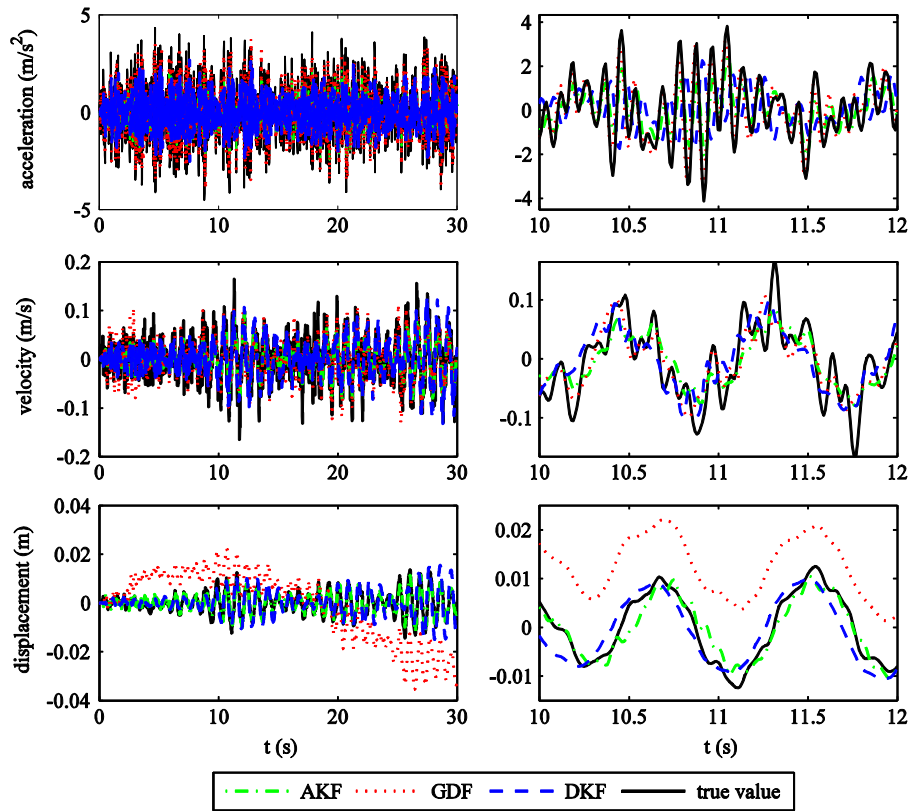


Fig. 13. Estimated displacement, velocity and acceleration time histories at DOF 5 of the shear building by AKF, GDF and DKF for the non-collocated white noise excitations.

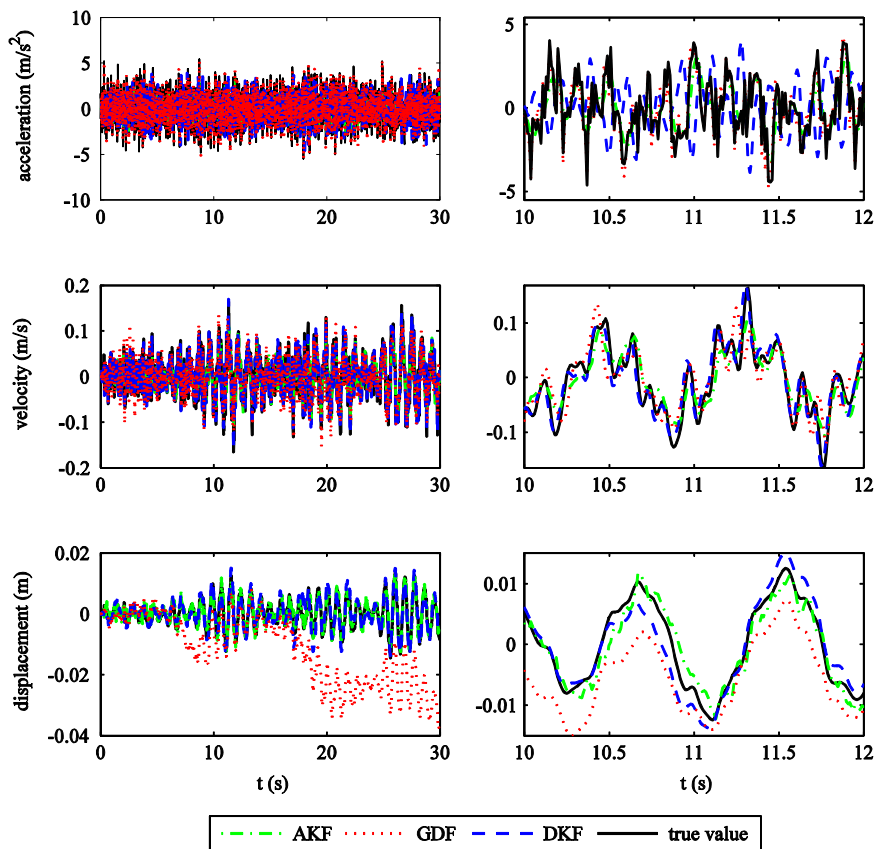


Fig. 14. Estimated displacement, velocity and acceleration time histories of the shear building at DOF 5 by AKF, GDF and DKF for the collocated and non-collocated white noise excitations.

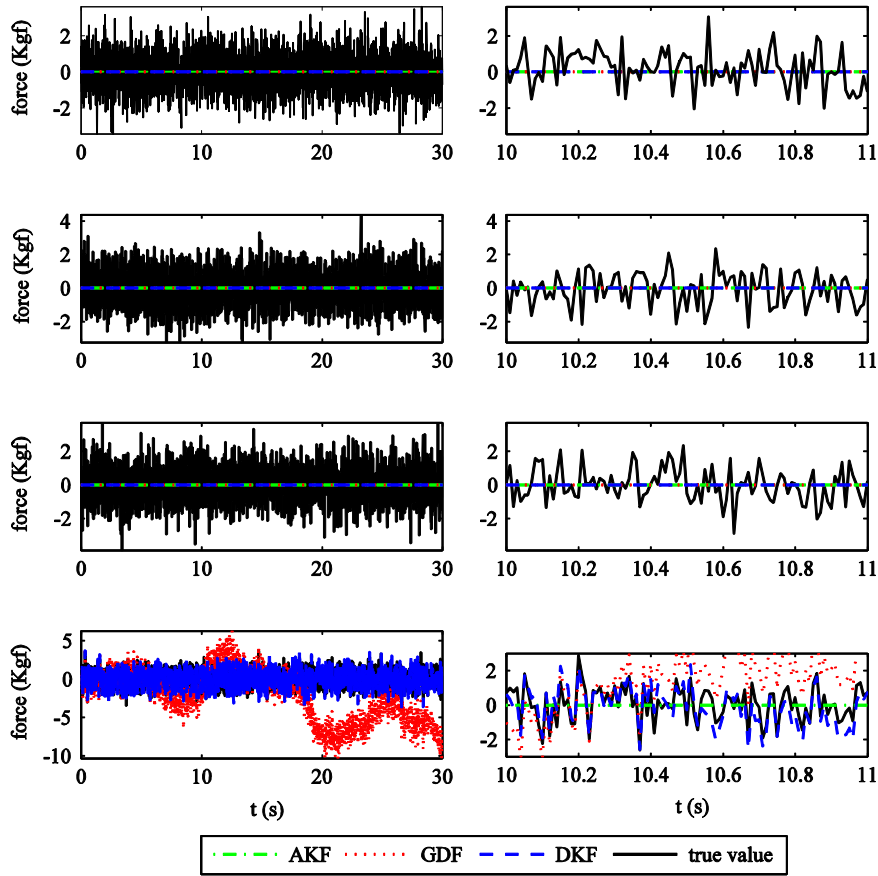


Fig. 15. Estimated input time histories of the shear building by AKF, GDF and DKF for the collocated and non-collocated white noise excitations for inputs applied to DOFs 2, 4, 6, and 8 from top to bottom, respectively.

filters. It is observed that, similar to previous case none of the methods provide estimates for inputs at non-collocated DOFs. However, for the collocated case the DKF provides reasonable estimates of the input, while the GDF estimates are affected with low frequency drift and AKF fails to estimate the input due to unobservability issues. Concerning the performance of the three filters in case of non-collocated input estimation, it is worth to remind that the input is estimated through application of the Kalman gain to the novelty. The Kalman gain, in the un-observed DOFs turns out to be a negligible amount. In turn, in the observed DOF, Kalman provides a large gain which leads to a physically wrong estimated input.

Before concluding the discussion on the simulated example, the effect of the modeling error on the accuracy of the estimates of the filter is investigated. In doing so, a reduced model of the shear building is considered, where the first 2 eigen modes of the structure are used to construct the model. A comparison of the constructed time histories of the reduced model versus that of the full model is demonstrated in Fig. 16, it is observed that the reduced model fails to fully capture the dynamics of the structure; therefore, the error covariances should be adjusted in the process equation. In Fig. 17 the results furnished by the three filters for estimation of the acceleration, velocity and displacement time histories of the DOF 5 are demonstrated. It is seen that, the use of the reduced model has not visibly downgraded the quality of the estimations.

At this point, it should be mentioned that for the deconvolution method as well after the input is reconstructed, assuming less inputs than outputs, a filter should then be applied for state estimation, given that the initial conditions are in reality unknown, and errors in both process and measurement exist. Therefore this is not a deterministic ode problem to solve for but rather a stochastic problem where traditionally the KF would be enforced for delivering an optimal state prediction under known input load. Instead, what is proposed herein, performs the filtering step in parallel.

5. Application to a field inspired test case: the Pirelli tower

As a case study, we investigate the capability of AKF, GDF and DKF in state estimation using unknown input by considering the Pirelli Tower in Milan, shown in Fig. 18. The building features 39 stories, and its total height is about 130 m. The plan dimensions of the standard floor are approximately $70 \times 20 \text{ m}^2$. The structural components of the building are entirely made of cast-in-place reinforced concrete. In particular, the lateral load resisting system is comprised of the four triangular cores, positioned at the edges of the plan and containing the staircases and technical compartments, by the four internal wall pillars, and by the central core, containing the lift compartments.

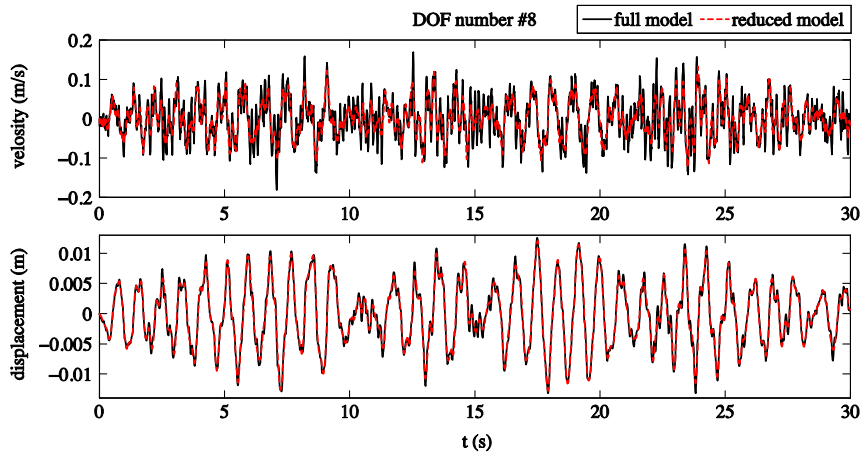


Fig. 16. Time histories of reduced and full model for the displacement and velocity of the DOF 8.

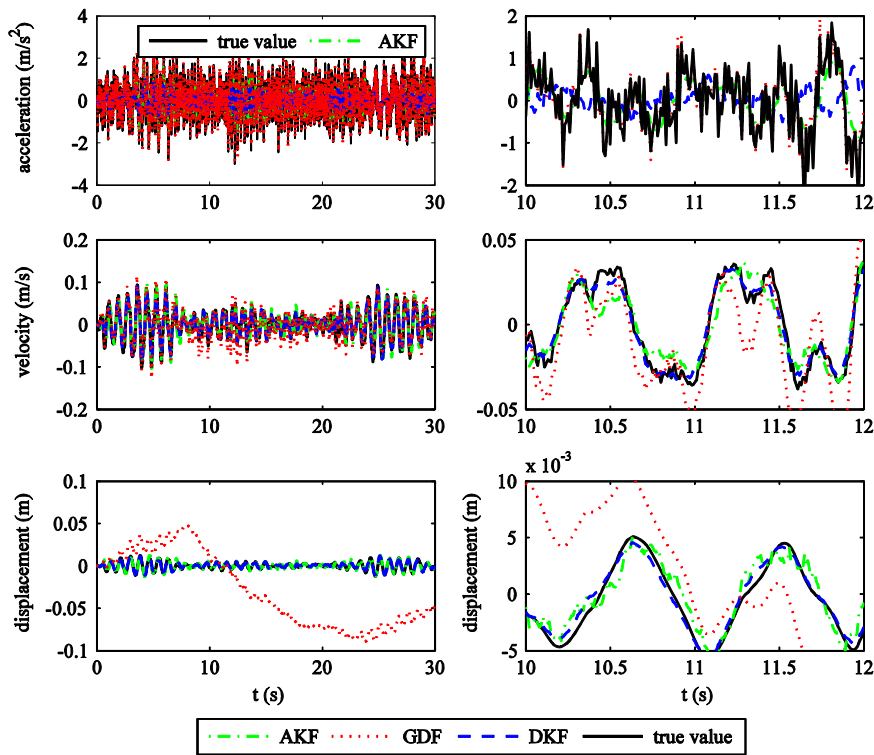


Fig. 17. Estimated displacement, velocity and acceleration time histories of the shear building at DOF 5 by AKF, GDF and DKF for the collocated and non-collocated white noise excitations, when 2 vibration modes are used in construction of the reduced model.

A three-dimensional finite element discretization of the whole building featuring $n = 6219$ DOFs have been used. In a relative frame moving with the basement of the tower, the undamped equations of motion of the structure read:

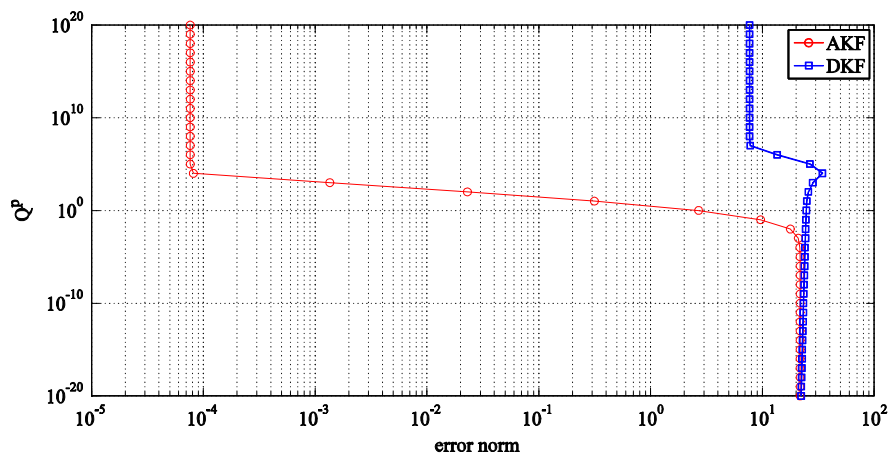
$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = -\mathbf{M}\mathbf{B} a(t) \quad (16)$$

where $\mathbf{u} \in \mathbb{R}^n$ and $a(t)$ signifies the seismic acceleration time history, whereas $\mathbf{B}$ is a Boolean matrix of appropriate dimension which indicates the points that the load is to be enforced. To simplify the problem, static condensation has been adopted to remove out of the simulation the vertical displacements of the floors. Moreover, the dynamics of the structure is analyzed in two dimensions; as lateral vibrations along the 70 m long axis are considered. Excluding the vertical degrees of freedom the equations of motion have a drastically lower order, where now $\mathbf{u} \in \mathbb{R}^n$, $n = 39$. The undamped natural frequencies of the system are reported in the Table 3. In the simulations, likewise the shear building case in the previous example, the modal damping has been assumed to be $\xi = 2\%$. At this point it needs to be noted that when tall buildings are excited by transient high frequency and pulse like inputs the mode superposition method may not be accurate; the delay in the effect of the applied load at the observed DOF may become significant.

Table 3

Lowest undamped natural frequencies of the building.

Vibration mode index	1	2	3	4	5	6	7	8	9	10	11	12	13
Natural frequency (Hz)	0.26	1.09	2.61	4.71	7.07	8.79	9.56	9.92	11.38	13.36	14.64	18.30	22.14

**Fig. 18.** The Pirelli Tower in Milan, Italy.**Fig. 19.** *L*-curve for the state and input identification of the Pirelli Tower with AKF and DKF using eight sensors.

The latter is due to the fact that when the dimension of the structure becomes large, the shear wave traveling time increases proportionally. Therefore, to alleviate the issues caused by this delay effect it is more efficient to use wave propagation methods for analyzing such structures as described in [39,40]. However, in this study the main objective is to accomplish a cross-comparison of the performance of the novel DKF to the previously developed AKF and GDF alternatives. To this end, the selected structure serves merely for visualization of a real case test example of higher dimensionality, hence the practical issues that arise when dealing with operational conditions are not addressed.

In what follows, the accelerogram of the May 6-1976 M_w 6.4, Friuli, January 17-1995 M_w 6.8 Kobe and May 18-1940 M_w 7.1 El Centro earthquakes are considered as the input to the structure. The acceleration records are scaled so that the peak ground acceleration in each case is 0.5 gm/s^2 . The $\mathbf{P}_0$, $\mathbf{Q}^\zeta$ and $\mathbf{R}$ are set to 10^{-20} , 10^{-20} and 10^{-1} , respectively. Fig. 19 shows the L -curve when the Friuli earthquake is used to shake the building. It is intuitively deduced that the value of the diagonal components of $\mathbf{Q}^p$ to be used in DKF and AKF could be set to 10^8 kgf^2 and 10^5 kgf^2 , respectively. Concerning the $\mathbf{P}_0^p$, the variations of the latter parameter does not result in an improved estimation, hence the presented results are relevant to same value used for $\mathbf{Q}^p$.

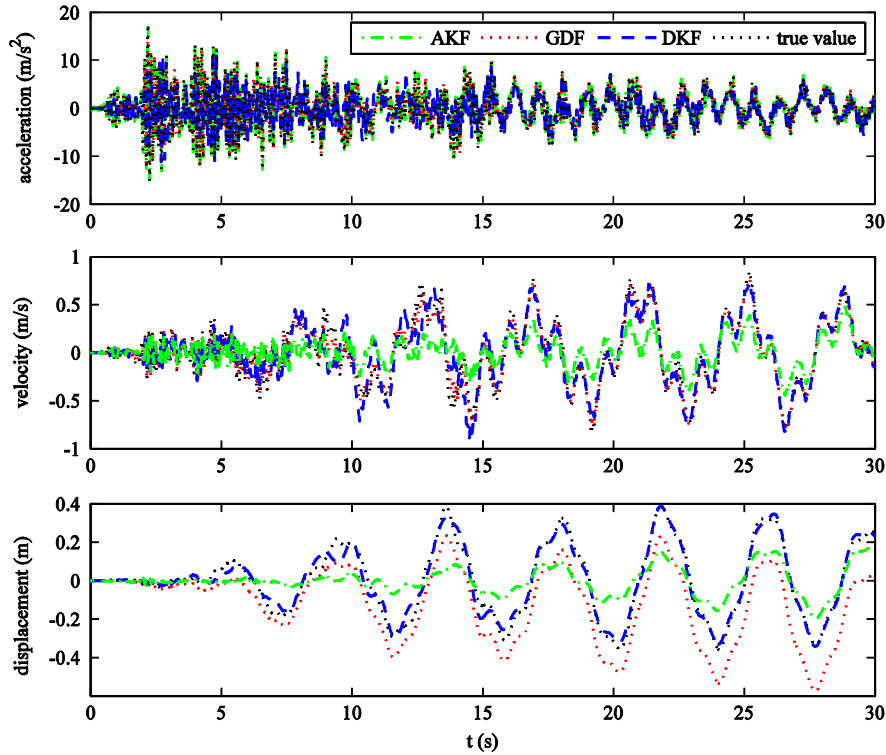


Fig. 20. Acceleration, velocity and displacement time history of the Pirelli Tower at DOF 25 estimated by the AKF, GDF and DKF for the Kobe Earthquake acceleration (top), velocity (middle), and displacement (bottom).

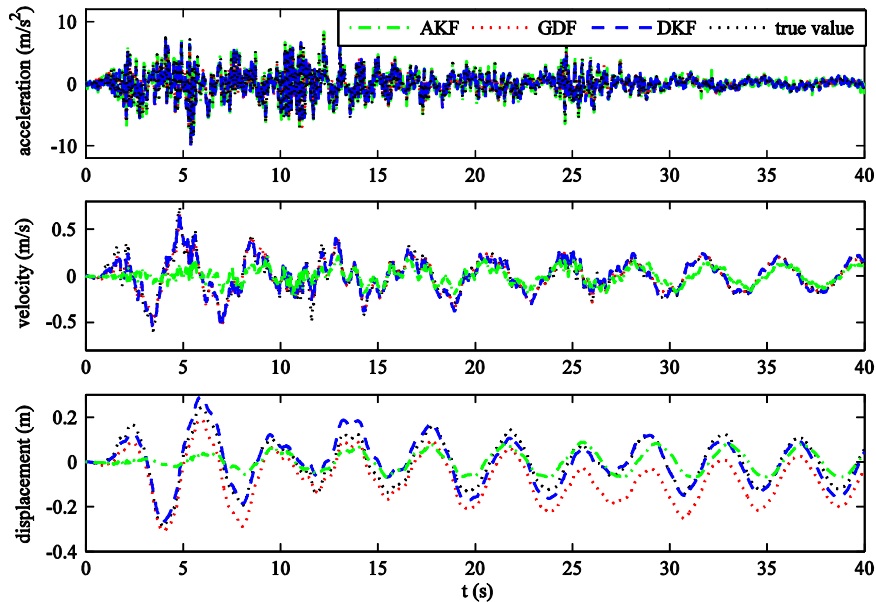


Fig. 21. Displacement, velocity and acceleration time histories at DOF 25 estimated by the AKF, GDF and DKF for the El Centro Earthquake acceleration (top), velocity (middle), and displacement (bottom).

It is assumed that there are 8 acceleration measurements at floor levels in the whole 39 floors of structure which starting from the last floor are evenly distributed over the height at floors 4, 9, 14, ..., 39. The standard deviation of the added noise signal is 0.31 m/s^2 , which leads to a noise-to-signal ratio that ranges from 1% up to 5% of the peak acceleration, depending on the considered DOF. The estimates made at the non-measured DOF 25 are presented herein, similar results are obtained at other measured and non-measured DOFs as well. The estimated time histories of the displacement, velocity and acceleration are shown in Fig. 20. It is seen that, dealing with DKF a reasonable agreement can be found between the estimated and true values of the sought-for states. The DKF and GDF estimates outperform those produced by AKF. Concerning the GDF, again the low frequency drifts in estimates are affecting the quality of the displacement results.

Next, we present the results of the analysis when the El Centro and Friuli earthquake records are used as input to the system. The values for the tuning parameters of the DKF are the same ones used for estimating the states and input by Kobe records. The estimated time histories of the displacement, velocity and acceleration are presented in Figs. 21 and 22, respectively. It is seen that results obtained by El Centro and Friuli earthquake records corroborate previous findings, and the variation in the seismic input does not alter trends found in Fig. 20.

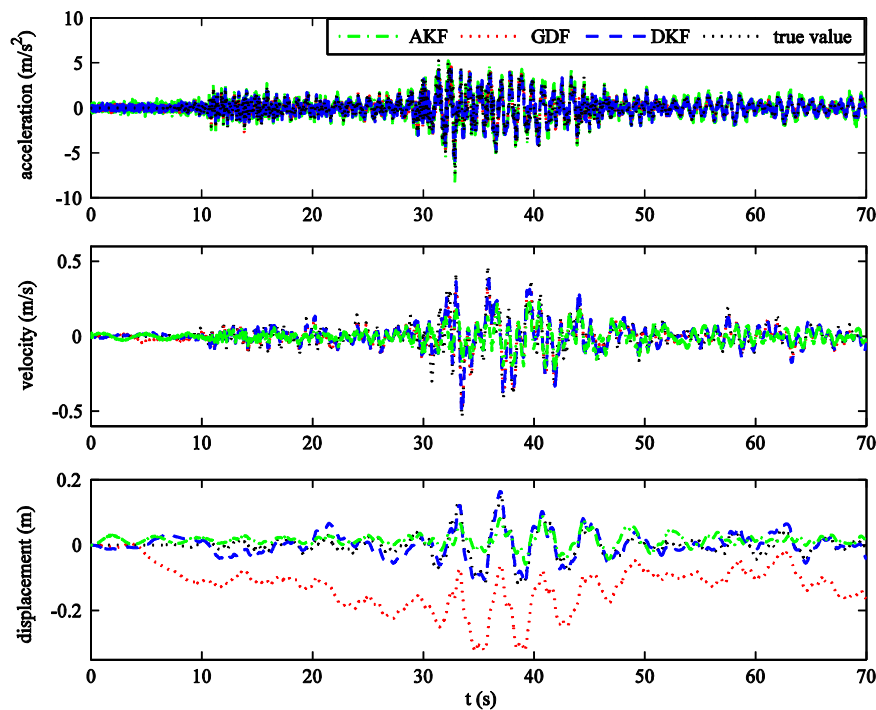


Fig. 22. Acceleration, velocity and displacement time histories at DOF 25 estimated by the AKF, GDF and DKF for the Friuli Earthquake acceleration (top), velocity (middle), and displacement (bottom).

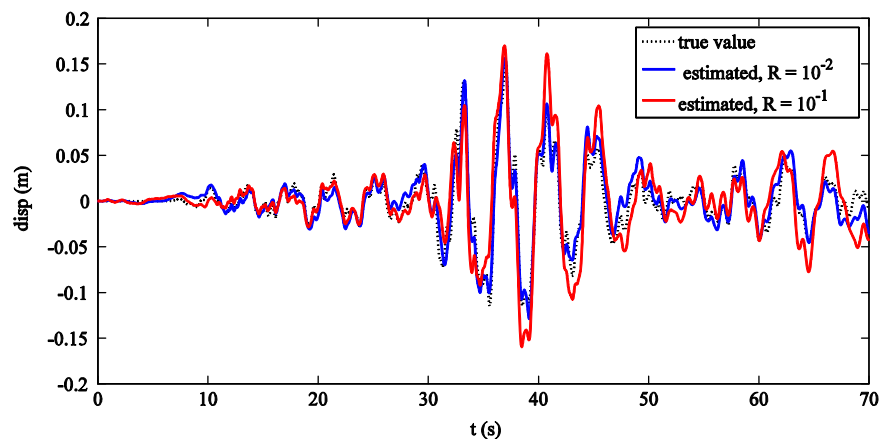


Fig. 23. Displacement time history estimated by the DKF in the case of the unknown input for different observation noise levels.

To study the effect of changes in the noise-to-signal ratio, Fig. 23 presents a comparison of the displacement time histories for two different acceleration noise levels. As expected, a more accurate sensor could provide significantly better estimates of the displacement time histories.

To make an assessment of the performances of DKF, GDF and AKF scheme when impact loading is applied to the structure, a similar impact load described in previous section is applied to the last floor. The impact load has a triangular time history: it starts at $t = 2$ s of the analysis, it takes 0.1 s to reach to its peak value which amounts 10^6 kgf and in another 0.1 s it descends back to zero. The spectrum of the applied force is shown in Fig. 24. The $\mathbf{P}_0$, $\mathbf{Q}^\zeta$ and $\mathbf{R}$ are set to 10^{-20} , 10^{-20} and 10^{-1} , respectively, whereas the $\mathbf{Q}^p$ used in DKF is set to 10^8 kgf. Concerning the $\mathbf{P}_0^p$, the variations of the latter parameter does not result in an improved estimation, hence the presented results are relevant to same value used for $\mathbf{Q}^p$.

In Figs. 25 and 26, the time histories of the displacement, velocity, acceleration and input force estimation versus their target value counterparts are confronted. It is inferred that the DKF outperforms the AKF and GDF in delivering accurate estimates of the displacement time history of the structure. However, concerning the acceleration and velocity time

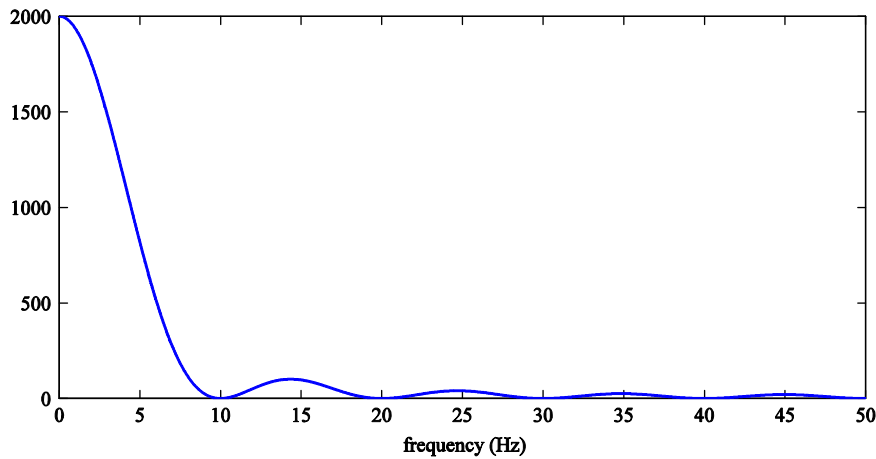


Fig. 24. Spectrum of the impulse force excitation applied to the Pirelli Tower.

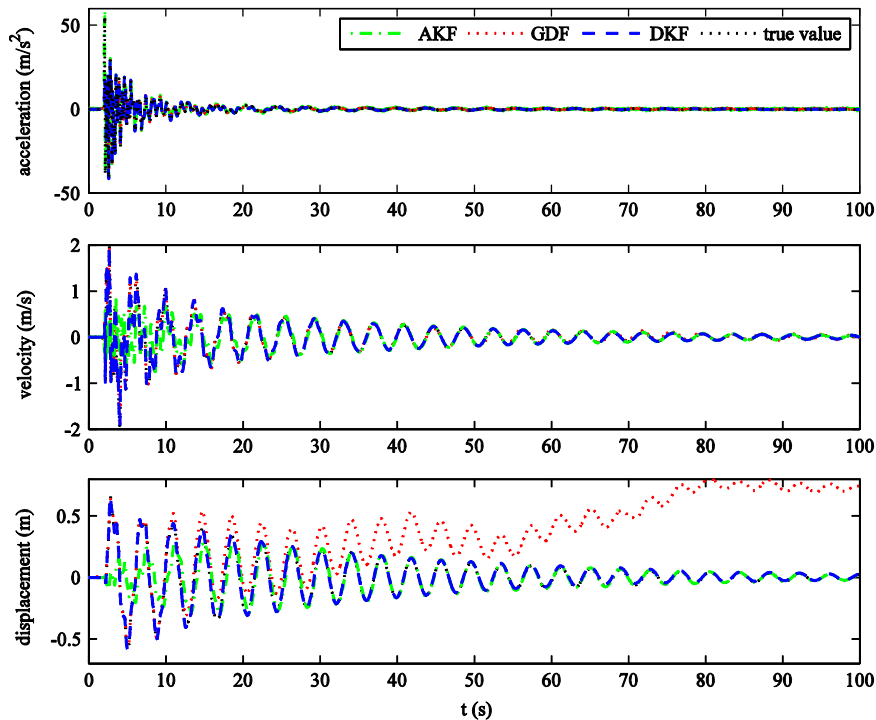


Fig. 25. Acceleration, velocity and displacement time histories at DOF 25 estimated by the AKF, GDF and DKF for the impact loading acceleration (top), velocity (middle), and displacement (bottom).

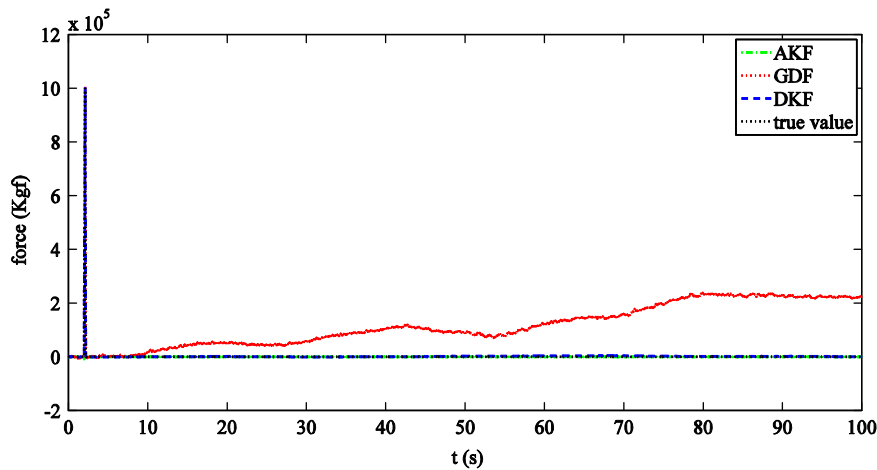


Fig. 26. Input force time history of the Pirelli Tower estimated by the AKF, GDF and DKF.

histories, GDF and DKF provide the same level of the accuracy of the estimates. Regarding the input force estimation, again the low frequency components affect the GDF performance. The AKF does not provide an estimate of the input force due to unobservability issues, while the DKF is suitably tracking the true value of the input force.

6. Conclusions

In this study, a dual implementation of the Kalman filter, namely the DKF, has been proposed to estimate the full states of a linear state-space model with unknown inputs. It is assumed that a limited number of noisy acceleration measurements are available. This data together with the known physical model of the system are incorporated into the DKF for accomplishing this objective. It has been demonstrated that the successive structure of the suggested filter prevents numerical problems attributed to un-observability and rank deficiency of the AKF. Additionally, it is revealed that the expert guess on the covariance of the unknown input provides a tool for avoiding the so-called drift effect in the estimated input force and displacements, observed in alternate implementations such as the GDF. The effectiveness and performance of the proposed method is ascertained via a simulated analysis carried out on a shear model of a building as well as the numerical model of the Pirelli Tower, a land mark skyscraper located at Milan, Italy. It is shown that the DKF outperforms the AKF and GDF in terms of the quality of the displacement estimates. It has been concluded that, by fine-tuning the covariance of the fictitious process noise of the unknown input, highly accurate estimates of displacements and velocities can be accomplished. The outcome of this study is important in estimating reliably the strain time history and thus tracing the fatigue damage accumulation in hotspot locations of structures using monitoring information from acceleration measurements.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.ymssp.2015.02.001>.

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